Species-Aware Module Review: Bacterial Translation Initiation in *Pseudomonas putida* KT2440

Target taxon: *Pseudomonas putida* KT2440 (PSEPK; NCBI taxon 160488; proteome UP000000556) **Nominal bucket:** KEGG `ppu03010` "Ribosome" (54 candidate genes) **Module under review:** Bacterial translation initiation (assembly of a translation-competent 70S initiation complex) **Date:** 2026-09-01

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1. Executive summary

- The **module is satisfiable** in *P. putida* KT2440. All three canonical initiation factors are encoded as single-copy genes with reviewed SwissProt entries: **IF-1** `infA` = **PP_4007** (P65117), **IF-2** `infB` = **PP_4712** (Q88DV7), **IF-3** `infC` = **PP_2466** (Q88K26).
- **Critical curation finding — bucket/module mismatch.** The resolved candidate bucket `kegg:ppu03010` "Ribosome" contains **only ribosomal proteins** (rps/*rpl*/rpm*) and none of the three initiation factors. KEGG map ppu03010 by definition enumerates the r-protein/rRNA parts of the 30S/50S subunits; the soluble initiation factors live in KEGG BRITE ko03012 "Translation factors" (≈ GO:0006413* translation initiation*). The 54 candidate genes therefore describe the ribosomal substrate / assembly platform**, not the module's active molecular players.
- Consequently **every module step maps to a real KT2440 gene, but those genes are outside the supplied candidate list.** Curation must pull IF genes from the translation-factor bucket, not from ppu03010.
- Evidence for the KT2440 IF genes is **UniProt homology transfer** (proteinExistence "Inferred from homology", annotation score 2.0), not direct in-species biochemistry. Assignments are nevertheless high-confidence because of sequence conservation, single-copy status, and conserved operon context (`thrS–infC–rpmI–rplT`, with rpmI = PP_2467 and rplT = PP_2468 already in the candidate list).

- Recommended module dispositions: **all three initiation steps → `covered`**, but flagged as `covered_offbucket` (genes not in ppu03010); the ppu03010 bucket itself → `module_needs_revision` for this module (wrong bucket, right organism).

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2. Target-organism pathway definition

In-scope biochemistry (this module). Assembly of the bacterial 70S initiation complex: early loading of IF2 and IF3 onto the 30S subunit; IF1-stabilized 30S pre-initiation complex (PIC); IF2-driven accommodation of initiator fMet-tRNA^fMet and 50S joining with GTP hydrolysis; and factor release yielding the mature 70S initiation complex. IF3's maintenance of free/newly-split 30S subunits is retained as the recycling→initiation interface.

Explicitly excluded (keep as separate buckets/maps): - Ribosome biogenesis / structure — the 54 ppu03010 r-proteins belong here (KEGG ppu03010, GO:0042254 assembly / GO:0005840 ribosome). They are the *platform*, not the initiation machinery. - **Initiator-tRNA charging and formylation** — methionyl-tRNA synthetase (`metG`) and methionyl-tRNA formyltransferase **`fmt` = PP_0067** (Q88RR2). fmt is present in KT2440 but is a boundary/excluded step. - **Elongation** — EF-Tu (`tufA` PP_0440 / `tufB` PP_0452), EF-Ts (`tsf` PP_1592), EF-G (`fusA` PP_0451 / `fusB` PP_4111). - **Ribosome recycling** — RRF (`frr` = PP_1594, Q88MH7) + EF-G; outside the module except via the IF3 interface above.

Alternate names / database definitions. IF1/IF2/IF3 = gene symbols `infA`/`infB`/`infC`. KEGG orthologs: K02518 (infA), K02519 (infB), K02520 (infC). KEGG BRITE ko03012 "Translation factors" is the correct grouping; the "Ribosome" pathway map (ko03010 / ppu03010) is a distinct, non-overlapping definition.

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3. Expected step model (generic module → KT2440 assignment)

#	Generic module step	Molecular player	KT2440 gene	Disposition
1	Early IF2/IF3 loading; free-30S maintenance	IF2 (early role), IF3	`infB` PP_4712; `infC` PP_2466	**covered** (off-bucket)
2	IF1-stabilized 30S pre-initiation complex	IF1	`infA` PP_4007	**covered** (off-bucket)
3	fMet-tRNA accommodation + 50S joining + GTP hydrolysis; factor release	IF2 (GTPase)	`infB` PP_4712	**covered** (off-bucket)
—	Initiator-tRNA formylation (boundary, excluded)	Fmt	`fmt` PP_0067	not_in_module (present)
—	Recycling→initiation interface	IF3 (retained) / RRF (excluded)	`infC` PP_2466 / `frr` PP_1594	interface covered by IF3

Known step order (IF2/IF3 → IF1 → IF2-GTPase/50S joining) is derived from *E. coli* real-time kinetics (Milón et al. 2012, P 22562136) and recent cryo-EM/fast-kinetics (Guerra et al. 2026, P 42427554). These are **not** KT2440 measurements; transfer to KT2440 is **strong** at the level of factor identity/role (deep conservation of IF1/IF2/IF3), but the precise assembly *sequence* should be treated as a favored *E. coli* route, not a proven KT2440 mechanism.

4. Candidate genes and evidence

4a. The module's true genes (NOT in the ppu03010 candidate list — must be added)

Gene	Locus	UniProt	Len	Evidence	Role	Caveats
`infA` (IF-1)	PP_4007	P65117 (reviewed)	72 aa	Homology-inferred; SwissProt	Stabilizes IF-2/ IF-3 on 30S, forms 30S PIC (step 2)	Small protein; single copy; no in-species assay
`infB` (IF-2)	PP_4712	Q88DV7 (reviewed)	846 aa	Homology-inferred; SwissProt; GTP-binding KW	Protects fMet-tRNA, promotes 30S binding + GTP hydrolysis for 70S (steps 1 & 3)	*E. coli* IF2 makes N-terminally truncated isoforms from internal starts — check for the same in KT2440 if isoform-level curation matters
`infC` (IF-3)	PP_2466	Q88K26 (reviewed)	183 aa	Homology-inferred; SwissProt; operon context	Free-30S maintenance / anti-association (step 1 + recycling interface)	May use non-AUG (AUU/AUG) autoregulatory start as in *E. coli* — verify start codon during fetch-gene review

4b. Candidate list (ppu03010) — role in this module

The 54 supplied genes are the **30S** (uS2–uS21, bS16/bS18/bS20/bS21, S1) and **50S** (uL1–uL30, bL9/bL17/…/bL36) ribosomal proteins. For the initiation module they are **context, not players**: they constitute the 30S and 50S subunits onto/into which the factors act. High-confidence, unambiguous r-protein annotations (standard bacterial nomenclature, one gene per protein). They should be marked as belonging to the **ribosome-structure module**, and used here only to confirm that both subunits are fully encoded (they are), so 30S availability and 50S joining are structurally possible.

Note two candidate loci provide operon evidence for the true module gene: **rpmI (PP_2467)** and **rplT (PP_2468)** are the downstream members of the `thrS–infC–rpmI–rplT` cluster, placing `infC` (PP_2466) immediately adjacent — independent support for the IF-3 assignment.

4c. Operon / synteny evidence (verified from KT2440 genome neighbors)

Gene-neighborhood context (UniProt loci) independently corroborates all three assignments and raises confidence beyond pure homology:

IF gene	Locus	Conserved operon context in KT2440	Note
`infC` (IF-3)	PP_2466	`thrS` (PP_2465)– **`infC`** – `rpmI` (PP_2467)– `rplT` (PP_2468)– `pheS` (PP_2469)	Classic `thrS-infC-rpmI-rplT`; rpmI/rplT are candidate-list genes
`infB` (IF-2)	PP_4712	`pnp` (4708)– `rpsO` (4709)– `truB` (4710)– `rbfA` (4711)– **`infB`** – `nusA` (4713)– `rimP` (4714)	Classic `rbfA-infB-nusA` / metY-nusA-infB macro-operon; rpsO is a candidate-list gene
`infA` (IF-1)	PP_4007	`aat` (4005)– `bpt` (4006)– **`infA`** – `clpA` (4008)– `clpS` (4009)	Standalone, as in *E. coli* (not in an r-protein operon)

Two candidate ppu03010 ribosomal genes physically flank two of the three module genes (**rpsO** next to infB; **rpmI/rplT** next to infC), so the wrong-bucket candidate list is nonetheless syntenically linked to the correct module genes — a useful curation cross-check.

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5. Gaps, ambiguities, and likely over-annotations

- **No gap in factor content.** IF1/IF2/IF3 all present, single-copy; no missing step.
- **Paralog ambiguity:** none for the IFs. (By contrast, elongation factors EF-Tu and EF-G *are* duplicated in KT2440 — tufA/tufB, fusA/fusB — but those are outside this module.)
- **Bucket-level over-scoping, not gene over-propagation.** The main annotation issue is that the *module was pointed at the wrong KEGG bucket*. There is no evidence of over-propagated IF annotation; the r-proteins are correctly r-proteins.
- **Loose "initiation factor" names to exclude (curation traps).** A proteome name/GO scan surfaces non-canonical hits that must NOT be pulled into the module:

- **PP_4196** (Q88FA2, "Translation initiation factor 2", 144 aa, *Predicted*) — far too short for real IF-2 (846 aa); over-propagated/loose name. **Exclude.**
- **PP_5605** (A0A140FWF6, "Secretion system X translation initiation factor", 168 aa, *Predicted*) — name tied to a secretion system; spurious for this module. **Exclude.**
- **PP_0566** `yciH` (Q88QC8, 123 aa, SUI1/eIF1-like domain, carries GO:0003743) — a genuine translation-*associated* SUI1-family protein, but **not** one of the canonical IF1/IF2/IF3; note as auxiliary only, do not use to satisfy any core step. Only PP_4007/PP_4712/PP_2466 should populate the module.
- **Evidence-quality caveat:** all three IF entries are homology-inferred (no direct KT2440 experiment). Genome-wide screens (Molina-Henares et al. 2010, P 20158506) did not report IF knockouts — consistent with IF1/IF2/IF3 being essential (non-recoverable), but this is inference, not a positive KT2440 result.
- **fMet-tRNA and metZ/metW tRNA genes** are not protein-coding candidates and won't appear in a proteome bucket; if the module requires the initiator tRNA as a component, that must be sourced from tRNA annotation, not KEGG proteome buckets.

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6. Module and GO-curation recommendations

1. **Reassign the bucket.** For "bacterial translation initiation," resolve to KEGG BRITE **ko03012 (translation factors)** / GO:0006413, not KEGG pathway map **ppu03010 (ribosome)**. Mark `kegg:ppu03010` as `module_needs_revision` *for this module* (correct organism, wrong bucket); keep ppu03010 as the bucket for a separate **ribosome-structure** module.
2. **Step dispositions:** Step 1 (IF2/IF3 loading) = **covered**; Step 2 (IF1 PIC) = **covered**; Step 3 (IF2 GTPase / 50S joining) = **covered**. Tag all as `covered_offbucket` because the supporting genes (PP_4007, PP_4712, PP_2466) are not in the supplied candidate set.
3. **Add genes to the module:** PP_4007 (infA), PP_4712 (infB), PP_2466 (infC). Optionally record boundary genes PP_0067 (fmt) and PP_1594 (frr) as `excluded/adjacent`.
4. **GO term requests:** none new required — GO:0003743 (translation initiation factor activity), GO:0006413 (translational initiation), GO:0003924 (GTPase activity, for IF2) already cover the module. Ensure PP_4712 carries GO:0003924 / GTP-binding.

5. **Module boundary check:** the generic boundaries (excluding elongation, termination, biogenesis, aminoacylation/formylation, RRF/EF-G recycling but retaining IF3 free-30S maintenance) are appropriate for KT2440 — no organism-specific boundary revision needed.

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7. Genes to promote to full `fetch-gene` review

1. **PP_4712** `infB` **(IF-2)** — highest priority: multidomain GTPase, module's catalytic hub; confirm GTPase motifs, GTP-binding annotation, and whether KT2440 produces IF2 isoforms (internal start sites) as in *E. coli*.
2. **PP_2466** `infC` **(IF-3)** — verify start codon (possible non-AUG autoregulatory start) and operon `thrS–infC–rpmI–rplT`; central to the recycling→initiation interface.
3. **PP_4007** `infA` **(IF-1)** — short protein; confirm annotation and (in)essentiality; low complexity but mechanistically required for step 2.

(Ribosomal-protein candidates do not need promotion for *this* module; they belong to the ribosome-structure module review.)

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8. Key references

- Milón P, Maracci C, Filonava L, Gualerzi CO, Rodnina MV. **Real-time assembly landscape of bacterial 30S translation initiation complex.** *Nat Struct Mol Biol* 2012. PMID **22562136**. — Establishes IF3/IF2-first, then IF1-locking assembly order (E. coli).
- Guerra J, et al. (Demo lab). **Molecular mechanism of IF1- and IF2-driven translation initiation in bacteria.** 2026. PMID **42427554**. — Cryo-EM/fast-kinetics of IF1/IF2 roles, GTP hydrolysis, Pi release, factor departure.
- Molina-Henares MA, et al. **Identification of conditionally essential genes for growth of *Pseudomonas putida* KT2440… genome-wide mutant library.** *Environ Microbiol* 2010. PMID **20158506**. — KT2440 mini-Tn5 knockout screen; IFs not recovered (consistent with essentiality).
- UniProtKB reviewed entries (taxon 160488): P65117 (IF1/PP_4007), Q88DV7 (IF2/PP_4712), Q88K26 (IF3/PP_2466), Q88RR2 (Fmt/PP_0067), Q88MH7 (RRF/PP_1594).

Evidence-transfer summary

Claim	Basis	Transfer strength to KT2440
IF1/IF2/IF3 present & single-copy in KT2440	UniProt reviewed, genome	**Direct** (in-species sequence)
IF functional roles per step	UniProt homology from *E. coli*	**Strong** (deep conservation)
IF gene identity correct (not mis-annotation)	UniProt homology + **conserved operon synteny** (thrS-infC-rpmI-rplT; rbfA-infB-nusA)	**Strong** (sequence + genomic context)
Assembly *sequence* (IF2/3→IF1→GTPase)	*E. coli* kinetics/cryo-EM	**Moderate** (mechanism not tested in KT2440)
IF essentiality in KT2440	Absence from Tn screen recoverables	**Weak/indirect**